Lab 4: Channel Estimation & Equalization

Summary

In Lab 4 you considered a communication system using a narrowband channel model. This scalar channel model does not reflect the frequency selective nature of most realistic wireless channels. This lab will introduce digital transmission and reception in a frequency selective channel, where multiple paths in the propagation environment create distortions in the transmitted. Correcting for distortions introduced by the channel requires more complicated receiver processing. In this lab you will implement additional features in the transmitter and receiver to mitigate channel distortions. The new transmitter, channel, and receiver structure are illustrated in Figure 1.

The main concepts explored in this lab are linear least squares estimation and linear equalization. After reviewing these concepts you will build the channel estimation and equalization blocks of the model in Figure 1. You will verify the functionality and correctness of your code in a simulator. Then, you will implement your code in lab using the USRP hardware to see how your code works over a real wireless link.

The goal of this lab is to understand the need for channel equalization and to understand basic algorithms for channel estimation and channel equalization. For the pre-lab submission, you have to turn in the three VIs described in Section 2 (*LLSE.vi*, *toeplitz.vi*, and *direct_equalizer.vi*). Furthermore, you have to submit the answers to questions in Section 2.

1 Background

In wireless communication systems, the signal encounters a effects in the propagation environment including reflections, scattering, diffraction, and path-loss. The different mechanisms of propagation create multiple propagation paths between the transmitter and receiver. For example, if there are two propagation paths, the receiver may observe the signal

$$z(t) \quad = \quad \alpha_0 e^{j\phi_0} x(t-\tau_0) + \alpha_1 e^{j\phi_1} x(t-\tau_1) + v(t), \qquad (1)$$

which consists of two different delayed, attenuated, and phase shifted versions of the signal $x(t)$. The delays τ_0 and τ_1 are determined by the propagation time of the paths, which would generally be different. If the difference $\tau_1 - \tau_0$ is significant (at least a fraction of a symbol period T) then the received signal will experience intersymbol interference. This can not be corrected using the

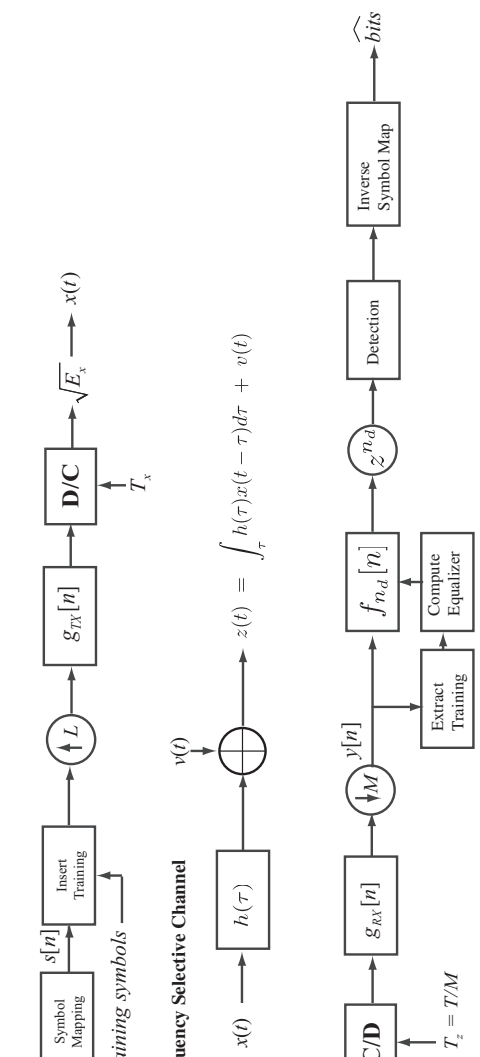


Figure 1: The system under consideration in this lab. The transmitter uses digital pulse-shaping and upsampling to create the transmit waveform. A known training sequence is inserted into the transmitted signal. The baseband signal encounters a frequency selective channel and additive white Gaussian noise at the receiver. The receiver uses digital pulse-shaping and downsampling, followed by linear equalization and detection, to find a good guess of the transmitted symbols. The linear equalizer is computed using the known training sequence in the received signal. An advance is applied after the equalization to correct for delays.

symbol synchronization techniques of Lab 3 since there is a sum of two different received signals.

In general, there may be many propagation paths between the transmitter and the receiver. A good generalization of Eq. (1) is

$$z(t) \quad = \quad \int_{\tau} h_e(\tau) x(t-\tau) d\tau + v(t), \tag{2}$$

where $h_e(\tau)$ is the baseband frequency selective channel. For complex pulse amplitude modulations, the channel distorts the pulse-shaping function. Consider the received signal after matched filtering and down-sampling. Using $\star$ for convolution define $h(t) := \sqrt{E_x} h_e(t) \star g_{\text{tx}}(t) \star g_{\text{rx}}(t) \star x(t)$ and let $h[n] = Th(nT)$ be the sampled version of the channel. Then

$$\begin{aligned} y[n] \quad &= \quad \sum_m s[m] h[n-m] + v[n] \\ &= \quad h[0] s[n] + \underbrace{\sum_{m \neq 0} s[m] h[n-m]}_{\text{intersymbol interference}} + v[n]. \end{aligned} \tag{3}$$

Unless $h(t)$ is a Nyquist pulse-shape, the second term in Eq. (3) will not be zero.

There are two main challenges associated with the received signal in Eq. (3).

1. The channel coefficients $\{h[\ell]\}$ create intersymbol interference. The solution we pursue in this lab is to augment the detection procedure with a *linear equalization* step.

2. The channel coefficients $\{h[\ell]\}$ are unknown to the receiver. The solution we pursue in this lab is to *estimate* the unknown channel coefficients and use them to determine the linear equalizer, or to estimate the coefficients of the linear equalizer directly.

The steps of estimation and equalization benefit from making some additional assumptions about the propagation channel. It is reasonable to assume that the propagation channel is causal and FIR (finite impulse response). It is causal because, naturally, the propagation channel can not predict the future. It is FIR because (i) there are no perfectly reflecting environments, and (ii) the signal energy decays as a function of distance between the transmitter and receiver. Essentially every time the signal reflects some of the energy passes through the reflector thus it looses energy. Additionally as the signal is propagating, it loses power as it spreads in the environment. Multipaths that are weak will fall below the noise threshold. With the FIR assumption, it

is common to assume that the composite channel $h(t)$ is also FIR thus Eq. (3) becomes

$$y[n] = \sum_{\ell=0}^{L} s[m]h[n-\ell] + v[n]. \quad (4)$$

The unknown channel coefficients are $\{h[\ell]\}_{\ell=0}^{L}$ where L is the order of the filter.

In this lab we will employ a linear equalizer. The goal of a linear equalizer is to find a filter that removes the effects of the channel. Let $\left\{f_{n_d}[l]\right\}_{l=0}^{L_f}$ be an FIR equalizer. The equalizer will be applied to the received signal so that

$$r[n] = \sum_{\ell=0}^{L_f} f_{n_d}[\ell]y[n-\ell] \approx \hat{s}[n-n_d], \quad (5)$$

where n_d is the equalizer delay and is generally a design parameter. Generally allowing $n_d > 0$ improves performance. The best equalizers consider several values of n_d and choose the best one.

1.1 Linear Least Squares

The main mathematical technique that will be used in this lab is known as linear least squares, which will be used both for estimating the channel and for computing the equalizer.

Let $\mathbf{A}$ denote a $N \times M$ matrix. The number of rows is given by N and the number of columns by M. If $N = M$ we say that the matrix is square. If $N > M$ we call the matrix tall and if $M > N$ we say that the matrix is fat. We use the notation $\mathbf{A}^T$ to denote the transpose of a matrix, $\mathbf{A}^*$ to denote the Hermitian or conjugate transpose, and $\mathbf{A}^c$ to denote the conjugate of the entries of $\mathbf{A}$. Let $\mathbf{b}$ be a $N \times 1$ dimension vector. The 2-norm of the vector is given by $\|\mathbf{a}\| = \sqrt{\sum_{n=1}^{N} |b_n|^2}$.

Consider a collection of vectors $\{\mathbf{x}_n\}_{n=1}^{K}$. We say that the vectors are linearly independent if there does not exist a set of nonzero weights $\{\alpha_n\}$ such that $\sum_n \alpha_n \mathbf{x}_n = \mathbf{0}$. In $\mathbb{C}^N$ and $\mathbb{R}^N$, at most N vectors can be linearly independent. We say that a square matrix $\mathbf{A}$ is invertible if the columns of $\mathbf{A}$ (or equivalently the rows) are linearly independent. If $\mathbf{A}$ is invertible, then a matrix inverse $\mathbf{A}^{-1}$ exists and satisfies the properties $\mathbf{A}\mathbf{A}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I}$. If $\mathbf{A}$ is tall, we say that $\mathbf{A}$ is full rank if the columns of $\mathbf{A}$ are linearly independent. Similarly, if $\mathbf{A}$ is a fat matrix, we say that it is full rank if the rows are linearly independent.

Consider a system of linear equations written in matrix form

$$\mathbf{Ax} = \mathbf{b} \tag{6}$$

where $\mathbf{A}$ is the known matrix of coefficients sometimes called the data, $\mathbf{x}$ is a vector of unknowns, and $\mathbf{b}$ is a known vector often called the observation vector. We suppose that $\mathbf{A}$ is full rank. First consider the case where $N = M$. Then the solution to Eq. (6) is $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$.

Now suppose that $N > M$. In this case $\mathbf{A}$ is tall and the system overdetermined in general. This means there are N equations but M unknowns, thus it is unlikely (except in special cases) that an exact solution exists. In this case we pursue an approximate solution known as least squares. Instead of solving Eq. (6) directly we instead propose to find the solution to the squared error

$$\min \|\mathbf{Ax} - \mathbf{b}\|^2. \tag{7}$$

Using matrix calculus, it can be shown that the solution to this problem assuming that $\mathbf{A}$ is full-rank is

$$\mathbf{x}_{LS} = (\mathbf{A}^*\mathbf{A})^{-1}\mathbf{A}^*\mathbf{b}. \tag{8}$$

Note that $\mathbf{A}^*\mathbf{A}$ is a square matrix and invertible because of the full-rank assumption. We refer to $\mathbf{x}_{LS}$ as the linear least square error (LLSE) solution to Eq. (6).

We use the squared error achieved by $\mathbf{x}_{LS}$ to measure the quality of the solution. With some calculus and simplifying, it can be shown that the least squares error achieved is given by

$$J(\mathbf{x}_{LS}) = \|\mathbf{Ax}_{LS} - \mathbf{b}\|^2 = \mathbf{x}^*\mathbf{A}^*(\mathbf{Ax} - \mathbf{b}) - \mathbf{b}^*(\mathbf{Ax} - \mathbf{b}) \tag{9}$$

$$= \mathbf{b}^*\mathbf{b} - \mathbf{b}^*\mathbf{Ax}. \tag{10}$$

Now you will use the least-squares solution for channel estimation and equalization.

1.2 Channel Estimation

There are several different criteria for designing an estimator including the maximum likelihood criterion, minimum mean squared error, and least squares. Of these, perhaps the least squares technique is the simplest thus we will discuss it here. The added advantage of the least squares estimator is that in AWGN the least squares estimator is also the maximum likelihood estimator.

Suppose that $\{t[n]\}_{n=0}^{N_t}$ is a known training sequence. This is a set of symbols that is known a priori. Suppose that the known training symbols